

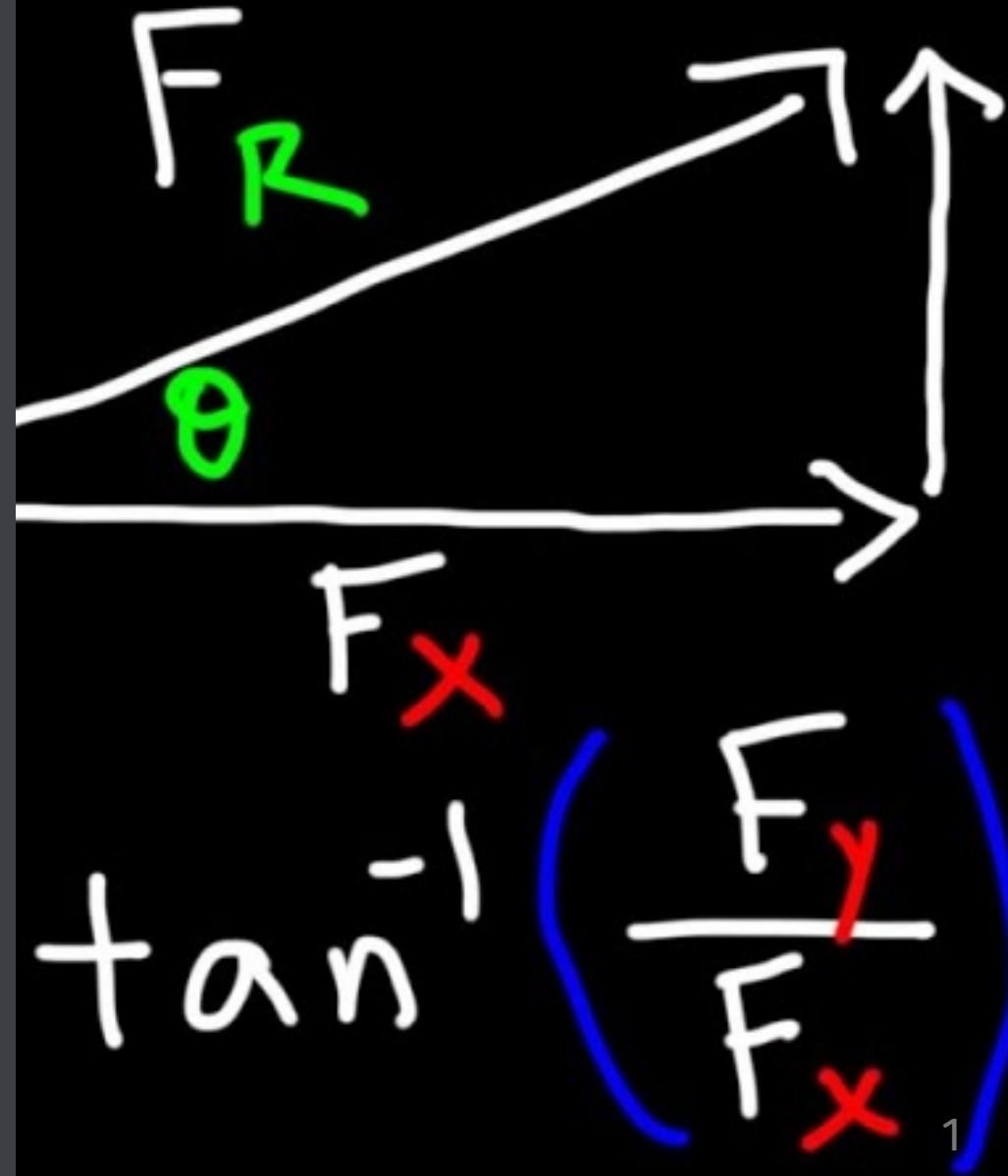
Day 03 - Vectors

$$\mathbf{r} = r_x \hat{x} + r_y \hat{y} + r_z \hat{z}$$

$$\mathbf{r} = r_x \hat{i} + r_y \hat{j} + r_z \hat{k}$$

$$\mathbf{r} = r_x \hat{e}_1 + r_y \hat{e}_2 + r_z \hat{e}_3$$

$$\mathbf{r} = \langle r_x, r_y, r_z \rangle$$



Announcements

- Homework 1 is due next Friday
- Help sessions will start next week
- Last class was not recorded properly

Goals for this week

Be able to answer the following questions.

- What is Classical Mechanics?
- How can we formulate it?
- What are the essential physics models for single particles?
- What mathematics do we need to get started?

Reminders from Day 02

- Classical Physics has existed for centuries
- Newton's Laws are one formulation of Classical Mechanics
- We can use Newton's Laws to describe the motion of particles
- Newton's Laws give rise to differential equations

Example: Ball Falling in 1D in Air

We derived the following differential equation for the motion of a ball falling in air:

$$m\ddot{y} = mg - bv - cv^2$$

We argued for low speeds, we neglect the v^2 term.

$$m\ddot{y} = mg - bv$$

We can instead write this differential equation for v :

$$\dot{v} = g - \frac{b}{m}v$$

Example: Ball Falling in 1D in Air

Is this integrable? **Yes!**

$$\frac{dv}{dt} = g - \frac{b}{m}v$$

$$\frac{dv}{g - \frac{b}{m}v} = dt$$

$$\int \frac{dv}{g - \frac{b}{m}v} = \int dt$$

Example: Ball Falling in 1D in Air

Let's perform a u -substitution:

Let $u = g - \frac{b}{m}v$, then $du = -\frac{b}{m}dv$

$$\int \frac{du}{u} = -\frac{m}{b} \int dt$$

$$\ln |u| = -\frac{m}{b}t + C$$

$$u = e^{-\frac{m}{b}t+C} = Ae^{-\frac{m}{b}t}$$

where $A = e^C$, a constant.

Example: Ball Falling in 1D in Air

In this limit, we are able to find an analytical solution for the velocity of the ball:

$$u = Ae^{-\frac{m}{b}t}$$
$$v = \frac{m}{b}g - Ae^{-\frac{m}{b}t}$$

If the ball starts at rest, $v(0) = 0$, then $A = \frac{m}{b}g$.

$$v = \frac{m}{b}g(1 - e^{-\frac{m}{b}t})$$

In the limit of large times, $t \rightarrow \infty$, $v \rightarrow \frac{m}{b}g$, the terminal velocity.

Vector Properties

Newtonian Mechanics is a vector theory. Here are a few mathematical properties of vectors:

- **Addition**: $\mathbf{A} + \mathbf{B} = (A_x + B_x)\hat{x} + (A_y + B_y)\hat{y} + (A_z + B_z)\hat{z}$
- **Scalar Multiplication**: $c\mathbf{A} = \langle cA_x, cA_y, cA_z \rangle$
- **Dot Product**: $\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z = AB \cos \theta$
- **Cross Product**: $\mathbf{A} \times \mathbf{B} = \langle A_y B_z - A_z B_y, A_z B_x - A_x B_z, A_x B_y - A_y B_x \rangle$
- **Unit Vectors**: $\hat{A} = \frac{\mathbf{A}}{|\mathbf{A}|} \quad |\hat{A}| = 1$

Creating a Generative Artificial Intelligence Policy

Underlying our work together is ensuring you develop a deep understanding of the physics and mathematics of Classical Mechanics.

We define **productivity** as the ability to use Generative AI to deepen your understanding of Classical Mechanics.

Take five minutes to answer the following for yourself:

1. What are ways that you think that AI can be used productively in our classroom?
2. What are ways that you think that AI can be used unproductively in our classroom?
3. What do you think are acceptable uses of AI in our classroom?
4. What do you think are unacceptable uses of AI in our classroom?
5. Once we define a policy, how should we collectively enforce it?

Creating a Generative Artificial Intelligence Policy

We define **productivity** as the ability to use Generative AI to deepen your understanding of Classical Mechanics.

Share your ideas at your table. Develop a consensus on the following:

1. What do y'all think are acceptable uses of AI in our classroom?
2. What do y'all think are unacceptable uses of AI in our classroom?
3. Once we define a policy, how should we collectively enforce it?

Add your answers to the form at the following link:

<https://forms.office.com/r/Bsh6ugKQ9Y>